

Software Design Document

The Factory Queue

Contents

The Factory Queue	1
1. Introduction.....	2
1.1. Product Scope	2
1.2. Requirements	2
1.3. Assumptions.....	2
2. Design	3-5
2.1. Class Diagram.....	3
2.2. Pseudo Code.....	4
2.3. Flow Charts.....	5
3. User Interface.....	6
4. Testing.....	7-10

1. Introduction

1.1. Product Scope

The purpose of this program is to help a factory manage its orders for Widgets.

1.2. Requirements

These are the requirements for this program.

- The program must have the following files: main.cpp, order.h, and order.cpp
- The order class must have the following 3 private data members: Customer Name (String), Order Number (int), and Number of Widgets (int)
- The order class must have the following constructor and 6 public methods: Order(customerName, orderNumber, numberOfWidgets), AddOrder(order), RemoveOrder(orderNum), ProcessNextOrder(), CheckNumberOfOrders(), PrintPendingOrders, ChangeNumWidgets(orderNum, numWidgets)
- The user should be able to perform the following 6 menu operations: add an order, remove an order, check how many orders are waiting to be processed, print a detailed list of all of the orders, process the next pending order, and change the number of widgets in an order
- This program must use a Queue
- The program must have at least 4 test cases

1.3. Assumptions

These are the assumptions made in the design and implementation of the program. It also covers the execution of the program.

- The program should be compiled without any errors.
- The menu serves as the user interface
- The user should use the menu options logically (i.e., adding order(s) THEN checking how many orders are in the queue instead of checking first then adding)
- The user should only enter valid integer types as a menu option; this program does not handle any menu options other than the ones provided
- The queue precedence is based on the sequence/time at which a user placed their order, with the back of the queue being orders placed more recently and the front of the queue being placed longer ago.
- We are free to decide how we can structure the order number. For example, it can be (#XXXXXX or #XXXX)
- “Processing an order” simply removes it from the queue and indicates it is processed with an associated output statement
- The queue in memory will only contain order numbers

2. Design

2.1. Class Diagram

Order
-string customerName -int orderNum -int numWidgets
+Order(customerName, orderNum, numWidgets) +void AddOrder(order) +void RemoveOrder(orderNum) +order ProcessNextOrder() +int CheckNumberOfOrders() +void PrintPendingOrders() +void ChangeNumWidgets(orderNum, numWidgets)

2.2. Pseudo Code

(This pseudo code emulates the main file to demonstrate program functionality)

Welcome the user to the widget factory ordering interactive system

While the user is not done using the program:

 Ask the user for a menu option

 If the menu option is 1:

 We will add an order by making a call to the AddOrder() method; this will prompt the user for their name and the number of widgets they plan to order, and we will use the random library to generate a random and unique 5-digit order number (the implementation of this method will detail the checking for order number uniqueness)

 Output “Done!” (may or may not be defined in the implementation)

 If the menu option is 2:

 We will remove an order by making a call to the RemoveOrder() method; this will ask the user for their order number and remove it from the Queue

 Output “Done!” (may or may not be defined in the implementation)

 If the menu option is 3:

 We will return the number of orders waiting to be processed by making a call to the CheckNumberOfOrders() method

 If the menu option is 4:

 Print a detailed list of all of the orders waiting to be processed by making a call to the PrintPendingOrders() method

 If the menu option is 5:

 We will process the next order in the queue by making a call to the ProcessNextOrder() method

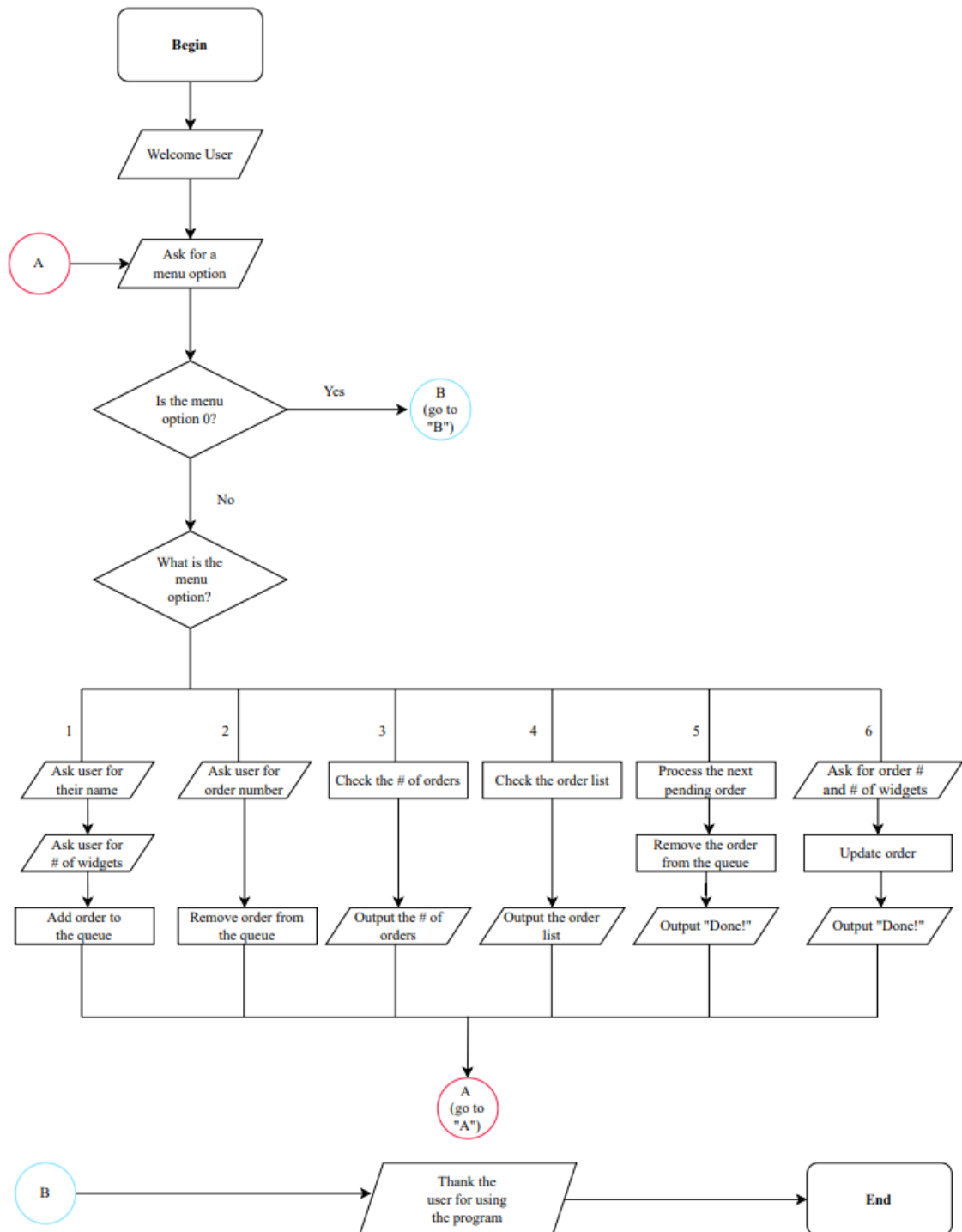
 Output “Done! Order #[orderNum] is processed!”
(may or may not be defined in the implementation)

 If the menu option is 6:

 We will appropriately change the number of widgets by making a call to the ChangeNumWidgets() method; this will ask the user for the newly specified number of widgets they would like to order and updates the order object to accurately reflect the order contents

Thank the user for using the program.

2.3. Flow Charts



3. User Interface

This program will be run on a command line prompt. When the program starts, it displays a welcome message and a menu with seven menu options that the user can perform (listed below for reference). The user should use the menu options logically and carefully (such as adding an order from the queue and then removing it).

The user may utilize the menu options as many times as they wish. Once the user enters menu option 0, the program will display a message thanking the user for using the program and thus ending it. The user may run the program as many times as they wish.

Menu Options:

0. End the program
1. Add an order (asks the user for their name and the number of widgets they would like)
2. Remove an order (asks the user for their order number)
3. Check the number of orders in the queue
4. Receive a list of the orders currently in the queue
5. Process the next pending order in the queue
6. Update their order by changing the number of widgets they would like (asks the user for their order number and the number of widgets they would like)

4. Testing

I will test if the program is correct and ensure that all the menu options function as intended using the following test cases. For our purposes, some menu options may be used in conjunction with another menu option for a given test case to demonstrate their functionality. Also, for the sake of brevity, I will not include the prompts asking the user for the order details in the expected output nor the output of a menu option if it is being used to demonstrate the functionality of a different menu option (unless it is needed for said demonstration, such as test case 5), and I will display the welcome and closing message only in the first and last test case, respectively.

Test Case 1

Purpose:

This is to test the functionality of the menu options 1 and 3.

Inputs:

1
Joaquin Ramirez
5
3

Expected Output:

Welcome to our ordering system for our state-of-the-art widgets! Please select one of the following menu options below!

Done. Order #[orderNum] added!

There is/are currently 1 order(s) in the queue.

Test Case 2

Purpose:

This is to test the functionality of menu option 2.

Inputs:

1
Joaquin Ramirez
5
2
[We will use whatever order number is generated]
3

Expected Output:

There is/are currently 0 order(s) in the queue.

Test Case 3

Purpose:

This is to test the functionality of the menu option 4 if there is atleast one order in the queue.

Inputs:

1
Joaquin Ramirez
8
1
Sarah Soto
3
1
Matthew Nguyen
5
1
Mina Pham
5
4

Expected Output:

The current orders in the queue are (from front of the queue to the back):

-Order #[orderNum]: Joaquin Ramirez, 8 widget(s)
-Order #[orderNum]: Sarah Soto, 3 widget(s)
-Order #[orderNum]: Matthew Nguyen, 5 widget(s)
-Order #[orderNum]: Mina Pham, 5 widget(s)

Test Case 4

Purpose:

This is to test the functionality of menu option 4 if there are no orders in the queue.

Inputs:

4

Expected Output:

There are currently no orders in the queue.

Test Case 5

Purpose:

This is to test the functionality of menu option 5 with atleast one order in the queue.

Inputs:

1
Joaquin Ramirez
8
1
Sarah Soto
3
5

Expected Output:

Done! Order #[orderNum] is processed.

The current orders in the queue are (from front of the queue to the back):

-Order #[orderNum]: Sarah Soto, 3 widget(s)

Test Case 6

Purpose:

This is to test the functionality of menu option 5 with no orders in the queue.

Inputs:

5

Expected Output:

No orders are currently in the queue.

Test Case 7

Purpose:

This is to test the functionality of menu option 6.

Inputs:

1
Joaquin Ramirez
8
6
[orderNum for Joaquin Ramirez]
10
4

Expected Output:

The current orders in the queue are (from front of the queue to the back):

-Order #[orderNum]: Joaquin Ramirez, 8 widget(s)

The current orders in the queue are (from front of the queue to the back):

-Order #[orderNum]: Joaquin Ramirez, 10 widget(s)

Test Case 8

Purpose:

This is to test the functionality of menu option 0.

Inputs:

1
Joaquin Ramirez
8
0

Expected Output:

Done. Order #[orderNum] added!

Thank you for using our ordering system. Have a widget-tastic day!